

PROGRAM 2:ADDING TWO NUMBERS

Xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:padding="16dp"/>
    <EditText
        android:id="@+id/num1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter first number"
        android:inputType="number"/>
    <EditText
        android:id="@+id/num2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="enter second number"
        android:inputType="number"/>
    <Button
        android:id="@+id/button"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="ADD"/>

    <TextView
        android:id="@+id/res"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

JAVA

```
package com.example.program2;

import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.EditText;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        EditText numb1=findViewById(R.id.num1) ;
        EditText numb2=findViewById(R.id.num2) ;
    }
}
```

```

Button butt=findViewById(R.id.button) ;
TextView result=findViewById(R.id.res) ;
butt.setOnClickListener(new View.OnClickListener()
{
    @Override
    public void onClick(View v) {
        int n1=Integer.parseInt(numb1.getText().toString());
        int n2=Integer.parseInt(numb2.getText().toString());
        int sum=n1+n2;
        result.setText("SUM="+sum);
    }
});
}
}

```

PROGRAM 3:ORIENTATION

XML:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/btn"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="potrait"/>
    <Button
        android:id="@+id/btn1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="layout"/>

</LinearLayout>

```

JAVA

```
package com.example.program3;

import android.content.pm.ActivityInfo;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button pot=findViewById(R.id.btn) ;
        Button lan=findViewById(R.id.btn1) ;
        pot.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {

                setRequestedOrientation(ActivityInfo.SCREEN_ORIENTATION_PORTRAIT);
            }
        });
        lan.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {

                setRequestedOrientation(ActivityInfo.SCREEN_ORIENTATION_LANDSCAPE );
            }
        });
    }
}
```

PROGRAM 4:PERSONAL DETAILS

XML CODE:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <EditText
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="enter your name:"
        android:id="@+id/name"/>
    <EditText
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="enter your mailid:"
        android:id="@+id/email"
        android:inputType="textEmailAddress"/>
    <EditText
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="enter your phone number:"
        android:id="@+id/phone">
```

```
        android:maxLength="10"
        android:inputType="phone"/>
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="enter your rollno:"
    android:id="@+id/rollno"/>
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="enter your course:"
    android:id="@+id/course"/>
<Button
    android:layout_width="match_parent"
```

```

        android:layout_height="wrap_content"
        android:text="Submit"
        android:id="@+id/btn"/>

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:id="@+id/displayinginfo"
    android:paddingTop="20dp"/>

</LinearLayout>

```

JAVA

```

package com.example.prog4;

import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        EditText name1=findViewById(R.id.name);
        EditText email1=findViewById(R.id.email);
        EditText phone1=findViewById(R.id.phone);
        EditText rollno1=findViewById(R.id.rollno);
        EditText course1=findViewById(R.id.course);
        Button btn1=findViewById(R.id.btn);
        TextView displayinginfo1=findViewById(R.id.displayinginfo);
        btn1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String name=name1.getText().toString();
                String email=email1.getText().toString();
                String rollno=rollno1.getText().toString();
                String phone=phone1.getText().toString();
                String course=course1.getText().toString();
                String display1="Name:"+name+"\nEmail Id:"+email+"\nPhone
no:"+phone+"\nRollno:"+rollno+"\nCourse"+course;
                displayinginfo1.setText(display1);
            }
        });
    }
}

```

PROGRAM 5

XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="countries available"
        android:textSize="20sp"
        android:textStyle="bold"
        android:textColor="@color/black"/>
    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:entries="@array/countries"/>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="select your favourite country"
        android:textSize="20sp"
        android:textStyle="bold"
        android:textColor="@color/black"/>
    <Spinner
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/spinner">
```

```

        android:entries="@array/countries"
        android:layout_marginTop="16dp"/>
<Button
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:id="@+id/button"
    android:text="Submit"
    android:layout_gravity="center"/>

</LinearLayout>

JAVA
package com.example.prog5;

import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Spinner;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Spinner spinner=findViewById(R.id.spinner) ;
        Button button=findViewById(R.id.button) ;
        button.setOnClickListener(new View.OnClickListener()
        {
            public void onClick(View View){
                String selected=spinner.getSelectedItem().toString();
                Toast.makeText(getApplicationContext(),"your favourite
country is"+selected,Toast.LENGTH_SHORT).show();
            }
        });
    }
}

```

STRINGS.XML

```

<resources>
    <string name="app_name">prog5</string>
    <string-array name="countries">
        <item>USA</item>
        <item>UK</item>
        <item>CANADA</item>
        <item>GERMANY</item>
        <item>FRANCE</item>
    </string-array>
</resources>

```

PROGRAM 6:

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">
    <EditText
        android:id="@+id/editText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter the text"/>
    <RadioGroup
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/radioGroup"
        android:layout_below="@id/editText"
        android:layout_marginTop="16dp"
        android:orientation="horizontal">
        <RadioButton
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/radioButton1"
            android:text=" I B.Sc CS"/>
        <RadioButton
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/radioButton2"
            android:text=" II B.Sc CS"/>
        <RadioButton
            android:layout_width="wrap_content"
```

```

        android:layout_height="wrap_content"
        android:id="@+id/radioButton3"
        android:text=" III B.Sc CS"/>
    </RadioGroup>
    <CheckBox
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/checkBox"
        android:layout_below="@id/editText"
        android:text="Check Me"
        android:layout_marginBottom="16dp"/>
    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/button"
        android:layout_marginTop="16dp"
        android:text="Show Text"/>

</LinearLayout>

```

JAVA:

```

package com.example.prog5;

import android.support.annotation.MainThread;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.EditText;
import android.widget.RadioGroup;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        EditText editText=findViewById(R.id.editText);
        CheckBox checkBox=findViewById(R.id.checkBox);
        RadioGroup radioGroup=findViewById(R.id.radioGroup);
        Button button=findViewById(R.id.button);
        button.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String text=editText.getText().toString();
                boolean isChecked=checkBox.isChecked();
                int SR=radioGroup.getCheckedRadioButtonId();
                String rt="";
                if (SR!=-1)
                {
                    RadioButton SRB= findViewById(SR);
                    rt=SRB.getText().toString();
                }
                String message=text+rt+(isChecked?"in PSGCAS":"unchecked");

                Toast.makeText(MainActivity.this,message,Toast.LENGTH_LONG).show();
            }
        });
    }
}

```

```

    }
    });
}

```

PROGRAM 7:

XML:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/qt"
        android:text="Questions will appear here"
        android:textSize="28sp"
        android:layout_alignParentTop="true"
        android:layout_centerHorizontal="true"/>

    <RadioGroup
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/ag"
        android:layout_below="@id/qt"
        android:layout_marginTop="16dp">

```

```

<RadioButton
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:id="@+id/o1"
    android:text="Option 1"/>
<RadioButton
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:id="@+id/o2"
    android:text="Option 2"/>
<RadioButton
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:id="@+id/o3"
    android:text="Option 3"/>
<RadioButton
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:id="@+id/o4"
    android:text="Option 4"/>
</RadioGroup>

<Button
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:id="@+id/sb"
    android:text="Submit Answer"
    android:layout_below="@id/ag"
    android:layout_marginTop="16dp"
    android:layout_centerHorizontal="true"/>

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:id="@+id/res_t"
    android:text="Result will appear here"
    android:textSize="18sp"/>
</LinearLayout>

```

JAVA:

```

package com.example.prog7v2;

import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        questiontextView=findViewById(R.id.qt);
        scoretextView=findViewById(R.id.res_t);
        answergroup=findViewById(R.id.ag);
        submitButton=findViewById(R.id.sb);

        loadQuestion();
    }
}

```

```

submitButton.setOnClickListener(v->{
    int selectedanswer=answergroup.getCheckedRadioButtonId();
    if (selectedanswer!=-1){
        int selectedindex=(selectedanswer==R.id.o1)?0:
            (selectedanswer==R.id.o2)?1:
            (selectedanswer==R.id.o3)?2:3;
        if (selectedindex==correctAnswer[currentindex]) score++;
        if(++currentindex<questions.length) loadQuestion();
        else{
            questiontextview.setText("Quiz Completed");
            scoretextview.setText("Final Score"+score);
            submitButton.setEnabled(false);
        }
    }
});
}

private void loadQuestion(){
    questiontextview.setText(questions[currentindex]);
    for (int i=0;i<4;i++){
        ((RadioButton) answergroup.getChildAt(i)).setText(options[currentindex][i]);
    }
    answergroup.clearCheck();
    scoretextview.setText("Score :"+score);
}

private TextView questiontextview,scoretextview;
private RadioGroup answergroup;
private Button submitButton;
private int currentindex=0,score=0;
private String[]questions={
    "What is the Capital of France?", "Which planet is known as Red
planet?", "Which is the Largest ocean on earth?"
};
private String[][] options={
    {"Paris", "London", "Berlin", "Medrid"},
    {"Earth", "Mars", "Jupiter", "Saturn"},
    {"Atlantic", "Indian", "Artic", "Pacific"}
};
private int[] correctAnswer={0,1,3};
}

```

PROGRAM 8:

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <LinearLayout
        android:id="@+id/linearLayout1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentLeft="true"
        android:layout_alignParentRight="true"
        android:layout_alignParentTop="true">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="ADDITION/SUBTRACTION"
            android:layout_gravity="center"
            android:textSize="20dp">

    </TextView>
</LinearLayout>

    <LinearLayout
        android:id="@+id/linearLayout2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentLeft="true"
        android:layout_alignParentRight="true"
        android:layout_below="@id/linearLayout1">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Enter no 1:"/>

        <EditText
            android:id="@+id/editText1"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="0.20"
            android:inputType="number">

    </EditText>
</LinearLayout>

    <LinearLayout
        android:id="@+id/linearLayout3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentLeft="true"
        android:layout_alignParentRight="true"
        android:layout_below="@id/linearLayout2">
```

```

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Enter no 2:"/>
        <EditText
            android:id="@+id/editText2"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="0.20"
            android:inputType="number">
    </EditText>
</LinearLayout>
    <LinearLayout
        android:id="@+id/linearlayout4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentLeft="true"
        android:layout_alignParentRight="true"
        android:layout_below="@id/linearLayout3">
        <Button
            android:id="@+id/button1"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="0.50"
            android:text="ADDITION"/>
        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/button2"
            android:layout_weight="0.50"
            android:text="SUBTRACTION"/>
        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/button3"
            android:layout_weight="0.50"
            android:text="CLEAR"/>
    </LinearLayout>
    <View
        android:id="@+id/linearlayout5"
        android:layout_width="fill_parent"
        android:layout_height="2px"
        android:background="#DDFFDD"/>
</RelativeLayout>

```

JAVA:

```

package com.example.prog8;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button; import android.widget.EditText;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
    EditText txtData1,txtData2;
    float num1,num2,result1,result2;
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button add = (Button) findViewById(R.id.button1);
        add.setOnClickListener(new OnClickListener() {

```

```

        public void onClick(View v) {
            try {
                txtData1 = (EditText) findViewById(R.id.editText1);
                txtData2 = (EditText) findViewById(R.id.editText2);
                num1=Float.parseFloat(txtData1.getText().toString());
                num2=Float.parseFloat(txtData2.getText().toString());
                result1=num1 + num2;
                Toast.makeText(getApplicationContext(), "ANSWER: " + result1,
Toast.LENGTH_SHORT).show();
            }
            catch (Exception e) {

Toast.makeText(getApplicationContext(),e.getMessage(),Toast.LENGTH_SHORT).show();
            }
        }
        Button sub= (Button) findViewById(R.id.button2);
        sub.setOnClickListener(new OnClickListener() {
            public void onClick(View v) {
                try {
                    txtData1 = (EditText) findViewById(R.id.editText1);
                    txtData2 = (EditText) findViewById(R.id.editText2);
                    num1 = Float.parseFloat(txtData1.getText().toString());
                    num2 = Float.parseFloat(txtData2.getText().toString());
                    result2=num1 - num2;
                    Toast.makeText(getApplicationContext(),"ANSWER:" + result2,
Toast.LENGTH_SHORT).show();
                }
                catch (Exception e) {

Toast.makeText(getApplicationContext(),e.getMessage(),Toast.LENGTH_SHORT).show();
                }
            }
        });
        Button clear = (Button) findViewById(R.id.button3);
        clear.setOnClickListener(new OnClickListener() {
            public void onClick(View v) {
                try {
                    txtData1.setText("");
                    txtData2.setText("");
                } catch (Exception e) {
                    Toast.makeText(getApplicationContext(), e.getMessage(),
Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}

```

PROGRAM 9:NOTIFICATION

XML:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <TextView
        android:id="@+id/messageTex
tView"
        android:layout_width="match
_parent"
        android:layout_height="wrap
_content" android:text="No
message yet"
        android:layout_centerHorizo
ntal="true"

```

```

        android:layout_marginTop="5
        0dp"
        android:textSize="18sp"/>
<Button
    android:id="@+id/receiveMessage
    Button"
    android:layout_width="match_par
    ent"
    android:layout_height="wrap_con
    tent" android:text="Receive
    Message"
    android:layout_below="@id/messa
    geTextView"
    android:layout_centerHorizontal
    ="true"
    android:layout_marginTop="20dp"
    />
</LinearLayout>\

```

JAVA:

```

package com.example.myapplication;

import
android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import
android.view.View;
import
android.widget.Button;
import
android.widget.TextView;
import
android.app.AlertDialog;
import android.content.DialogInterface;

public class MainActivity extends
    AppCompatActivity { private TextView
    messageTextView;
    private Button receiveMessageButton;

    @Override
    protected void onCreate(Bundle savedInstanceState)
    { super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    messageTextView=findViewById(R.id.messageTextView);

```

```

        receiveMessageButton=findViewById(R.id.receiveMessageButton);
        receiveMessageButton.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v) {
                String message="You have received a new message";
                messageTextView.setText(message);
                new AlertDialog.Builder(MainActivity.this)
                    .setTitle("New messge")
                    .setMessage(message)
                    .setPositiveButton("OK", new
                        DialogInterface.OnClickListener() { @Override
                            public void onClick(DialogInterface dialog, int
                                which) { dialog.dismiss();
                            }
                        })
                    .show();
            }
        });
    }
}

```

PROGRAM 10:

MyView.java:

```

package
com.example.program10;
import
android.content.Context;
import
android.graphics.Canvas;
import
android.graphics.Color;
import
android.graphics.Paint;
import
android.util.AttributeSet
; import
android.view.View;
public class MyView extends View
{
    private Paint
    paint; private
    float circleX=100;

```

```

private float circleY=300;
public MyView(Context context, AttributeSet attrs)
{
    super(context, attrs);
    paint=new
    Paint();
    paint.setColor(Color
    or.BLUE);
    paint.setStyle(Paint
    Style.FILL);
    paint.setAntiAlias
    (true);
}
protected void onDraw(Canvas canvas)
{
    super.onDraw(canvas);
    canvas.drawCircle(circleX, circleY, 80, paint);
}
public void setCircleX(float x)
{
    this.circleX=
    x;
    invalidate();
}
public void setCircleY(float y)
{
    this.circleY=
    y;
    invalidate();
}
}

```

XML:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#FFF8E1"
    tools:context=".MainActivity">
    <com.example.program10.MyView
        android:id="@+id/myView"
        android:layout_width="match_p
        arent"
        android:layout_height="match_
        parent"/>
    <Button
        android:id="@+id/btn
        Animate"
        android:layout_width="wrap_
        content"
        android:layout_height="wrap
        _content"
        android:text="Start
        Animation"
        android:layout_centerHorizo
        ntal="true"
        android:layout_alignParentB
        ottom="true"
        android:layout_marginBottom
        ="40dp"/>

```

```
</RelativeLayout>
```

MainActivity.java

```
package com.example.program10;

import
android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import
android.animation.ObjectAnimator
; import android.widget.Button;

public class MainActivity extends
AppCompatActivity { private MyView
myView;
@Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        myView=findViewById(R.id.myView);

        Button btnAnimate=findViewById(R.id.btnAnimate);
        btnAnimate.setOnClickListener(v ->{
ObjectAnimator
    animator=ObjectAnimator.ofFloat(myView,"CircleY",100,90
0); animator.setDuration(2000);
    animator.setRepeatCount(3);
    animator.setRepeatMode(ObjectAnimator.REVERSE);
    animator.start();
        });
    }
}
```

PROGRAM11:NOTIFICATION FROM ONE DEVICE TO ANOTHER

ACTIVITY_MAIN.XML:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/messageEditText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter message"
        android:inputType="text"/>
    <Button
        android:id="@+id/sendButton"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Send Message"
        android:layout_marginTop="12dp"/>
</LinearLayout>
```

MainActivity.java:

```
package com.example.program11;

import
    android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import
    android.content.Intent;
import
    android.view.View;
import
    android.widget.Button;
import
    android.widget.EditText;
;

public class MainActivity extends
    AppCompatActivity { @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        EditText
            messageEditText=findViewById(R.id.messageEditText); Button
            sendButton=findViewById(R.id.sendButton);
        sendButton.setOnClickListener(new
            View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String message=messageEditText.getText().toString();
```

```
Intent intent=new Intent(MainActivity.this,SecondActivity.class);
intent.putExtra("message_Key",message);
startActivity(intent);
}
```

```

    });
}
}

```

SecondActivity.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".SecondActivity">
    <TextView
        android:id="@+id/recievedMessage"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Received Message"
        android:textSize="18sp"/>

</LinearLayout>

```

SecondActivity.java

```

package com.example.program11;

import
    android.support.v4.app.NotificationCompat;
import
    android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.os.Build;
import
    android.app.NotificationChannel;
; import
    android.app.NotificationManager;
; import
    android.content.Context;
import android.widget.TextView;

public class SecondActivity extends
    AppCompatActivity { private TextView
        recievedMessageTextView;
    private final String CHANNEL_ID="message_channel";

    @Override
        protected void onCreate(Bundle
            savedInstanceState) {
            super.onCreate(savedInstanceState);
            setContentView(R.layout.activity_second);
            recievedMessageTextView=findViewById(R.id.recievedMessage); String
                message=getIntent().getStringExtra("message_key");
            recievedMessageTextView.setText(message);
            showNotification(message);
        }
        private void showNotification(String message)
        {
            NotificationManager notificationmanger=(NotificationManager)
                getSystemService(Context.NOTIFICATION_SERVICE);

```

```
if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O)
```

```

{
    NotificationChannel channel=new NotificationChannel(CHANNEL_ID,"Message
Notifications",NotificationManager.IMPORTANCE_DEFAULT);
    notificationmanger.createNotificationChannel(channel);
}

    NotificationCompat.Builder builder=new
    NotificationCompat.Builder(this,CHANNEL_ID)
    .setSmallIcon(android.R.drawable.ic_dialog_info)
    .setContentTitle("New Message Recieved")
    .setContentText(message)
    .setPriority(NotificationCompat.PRIORITY_DEFAULT);
    notificationmanger.notify(1,builder.build());
}
}

```

PROGRAM 12:MOBILE GAME APPLICATION

XML:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android
    " xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ImageView
        android:id="@
        +id/ball"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"

```

```

        android:src="@drawable/ball"/>
<TextView
    android:id="@+id/scoreText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Score:0"
    android:textSize="24sp"
    android:layout_alignParentTop="true"
    android:layout_centerHorizontal="true"/>

</RelativeLayout>

```

JAVA:

```

package com.example.program12;

import
    android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import
    android.view.MotionEvent;
import
    android.annotation.SuppressLint;
import android.view.View;
import
    android.widget.ImageView;
import
    android.widget.TextView;
import java.util.Random;
public class MainActivity extends
    AppCompatActivity { private ImageView
    ball;
    private Random random = new
    Random(); private int score = 0;
    private
    TextView scoreText;
    @SuppressWarnings("ClickableViewAccessibility") @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        ball = findViewById(R.id.ball);
        scoreText =
        findViewById(R.id.scoreText);
        ball.setOnClickListener((v,
        event) -> {
        if (event.getAction() ==
        MotionEvent.ACTION_DOWN) { score++;
        scoreText.setText(String.valueOf("Score: " +
        score)); moveBall();
        return true;
        }
        return false;
        });
        moveBall();
    }

    private void moveBall() {
        int screenWidth =

```

```
getResources().getDisplayMetrics().widthPixel  
s; int screenHeight =  
getResources().getDisplayMetrics().heightPixe  
ls;
```

```

        int x = random.nextInt(screenWidth -
        ball.getWidth()); int y =
        random.nextInt(screenHeight -
        ball.getHeight()); ball.setX(x);
        ball.setY(y);
    }
}

```

BALL.XML

```

<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android">
    <solid android:color="#FF0000"/>
    <size
        android:width="50dp"
        android:height="50dp"/>
    <corners android:radius="25dp"/>
</shape>

```

PROGRAM 13:AUDIO MODE(NORMAL,SILENT,VIBRATE)

XML:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/androi
d" xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"

```

```

        android:padding="24dp"
        tools:context=".MainActivity">
        <TextView
            android:id="@+id/m
            odetext"
            android:layout_width="wrap_
            content"
            android:layout_height="wrap
            _content"
            android:text="Current
            mode:"
            android:textSize="18sp"
            android:layout_marginBottom
            ="24dp"/>
        <Button
            android:layout_width="wrap_
            content"
            android:layout_height="wrap
            _content"
            android:id="@+id/normalbtn"
            android:text="Normal
            Mode"/>
        <Button
            android:layout_width="wrap_
            content"
            android:layout_height="wrap
            _content"
            android:id="@+id/silentbtn"
            android:text="Silent Mode"
            android:layout_marginTop="1
            6dp"/>
        <Button
            android:layout_width="wrap_
            content"
            android:layout_height="wrap
            _content"
            android:id="@+id/vibratebtn
            " android:text="Vibrate
            Mode"
            android:layout_marginTop="1
            6dp"/>
    </LinearLayout>

```

JAVA:

```

package
com.example.program13;
import
android.media.AudioManager;
import
android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import
android.widget.Button;
import
android.widget.Switch;
import
android.widget.TextView;
public class MainActivity extends
    AppCompatActivity { AudioManager
    audioManager;
    TextView
    modeText;
    @Override
    protected void onCreate(Bundle savedInstanceState) {

```

```
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
audioManager = (AudioManager)
getSystemService(AUDIO_SERVICE); modeText =
findViewById(R.id.modetext);
Button nbtn =
findViewById(R.id.normalbtn)
; Button sbtn =
findViewById(R.id.silentbtn)
; Button vbtn =
findViewById(R.id.vibratebtn
); updateModeText();
nbtn.setOnClickListener(v->{
```

```

        audioManager.setRingerMode(AudioManager.RINGER_MODE_
NORMAL); updateModeText();
    });

    sbtn.setOnClickListener(v->{
        audioManager.setRingerMode(AudioManager.RINGER_MODE_SIL
ENT); updateModeText();
    });
    vbtn.setOnClickListener(v->{
        audioManager.setRingerMode(AudioManager.RINGER_MODE_VIB
RATE); updateModeText();
    });
}
private void updateModeText() {
    int mode =
        audioManager.getRingerMode()
        ; String modestr;
    switch (mode) {
    case
        AudioManager.RINGER_MODE_NORMAL:
        modestr = "NORMAL";
        break;
    case
        AudioManager.RINGER_MODE_SIL
ENT: modestr = "SILENT";
        break;
    case
        AudioManager.RINGER_MODE_VIBRATE:
        modestr = "VIBRATE";
        breake
        k;
    default
    t:
        modestr =
            "UNKNOWN"; break;
    }
    modeText.setText("Current mode: " + modestr);
}
}

```

MANIFEST.XML:

```

<uses-permission android:name="android.permission.MODIFY_AUDIO_SETTINGS"/>

```

PROGRAM 14:

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <EditText
        android:id="@+id/
        emailto"
        android:layout_width="match
        _parent"
        android:layout_height="wrap
        _content"
        android:hint="Recipient
        Email"
        android:inputType="textEma
        ilAddress" />
```

```

<EditText
    android:id="@+id/emailsubject"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Subject"
    android:inputType="text"
/>
<EditText
    android:id="@+id/emailmessage"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Message"
    android:inputType="textMultiLine"
    android:minLines="3" />
<Button
    android:id="@+id/sendemailbutton"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Send Email" />
</LinearLayout>

```

JAVA:

```

package com.example.program14;

import
android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import
android.support.v7.app.AppCompatActivity;
import android.view.View;
import
android.widget.Button;
import
android.widget.EditText
; import
android.widget.Toast;
public class MainActivity extends
AppCompatActivity { private EditText
emailTo, emailSubject, emailMessage;
private Button sendEmailButton;
@Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        emailTo =
        findViewById(R.id.emailto);
        emailSubject =
        findViewById(R.id.emailsubject);
        emailMessage =
        findViewById(R.id.emailmessage);
        sendEmailButton = findViewById(R.id.sendemailbutton);
        sendEmailButton.setOnClickListener(new View.OnClickListener() {
@Override

```

```
        public void
            onClick(View v) {
                sendEmail();
            }
        });
    }
    private void sendEmail() {
        String to = emailTo.getText().toString();
```

```

        String subject =
            emailSubject.getText().toString();
        String message =
            emailMessage.getText().toString();
        Intent emailIntent = new
            Intent(Intent.ACTION_SEND);
        emailIntent.setType("message/rfc822"
            );
        emailIntent.putExtra(Intent.EXTRA_EMAIL, new
            String[]{to});
        emailIntent.putExtra(Intent.EXTRA_SUBJECT,
            subject); emailIntent.putExtra(Intent.EXTRA_TEXT,
            message);
        try {
            startActivity(Intent.createChooser(emailIntent, "Send email using"));
        } catch (android.content.ActivityNotFoundException ex) {
            Toast.makeText(this, "No email client installed.",
                Toast.LENGTH_SHORT).show();
        }
    }
}

```

PROGRAM 15:FIREBASE

Main_activity.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/
        android"
    xmlns:app="http://schemas.android.com/apk/res-
        auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"

```

```

        android:layout_width="match_paren
t"
        android:layout_height="match_pare
nt" tools:context=".MainActivity"
        android:orientation="vertical"
        android:layout_margin="100px">
        <EditText
            android:layout_width="match_p
arent"
            android:layout_height="wrap_c
ontent" android:hint="Enter
Username"
            android:id="@+id/username"
            android:inputType="textEmailA
ddress"/>
        <EditText
            android:layout_width="match
_parent"
            android:layout_height="wrap
_content"
            android:hint="Enter
Password"
            android:id="@+id/password"
            android:inputType="textPass
word"/>
        <Button
            android:layout_width="match
_parent"
            android:layout_height="wrap
_content"
            android:id="@+id/btnlogin"
            android:text="Login"/>
    </LinearLayout>

```

ActivityMain.java:

```

package com.example.prg15;

import
android.content.Intent;
import
android.os.Bundle;
import
android.view.View;
import
android.widget.Button;
import
android.widget.EditText
; import
android.widget.Toast;
import androidx.annotation.NonNull;
import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import
com.google.android.gms.tasks.OnCompleteListener;
import com.google.android.gms.tasks.Task;
import
com.google.firebase.auth.AuthResult
; import
com.google.firebase.auth.FirebaseAuth;

public class MainActivity extends
AppCompatActivity { @Override

```

```
protected void onCreate(Bundle savedInstanceState) {  
    super.onCreate(savedInstanceState);  
    setContentView(R.layout.activity_main);  
  
    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main),  
        (v, insets) -> { Insets systemBars =  
            insets.getInsets(WindowInsetsCompat.Type.systemBars());  
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,  
                systemBars.bottom);  
        });  
}
```

```

        return insets;
    });

    EditText username =
        findViewById(R.id.username);
    EditText password =
        findViewById(R.id.password);
    Button btnlogin =
        findViewById(R.id.btnlogin);

    FirebaseAuth mAuth = FirebaseAuth.getInstance();

    btnlogin.setOnClickListener(new

        View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String email =
                    username.getText().toString().trim();
                String pass =
                    password.getText().toString().trim();

                if (email.isEmpty() || pass.isEmpty()) {
                    Toast.makeText(MainActivity.this, "Fields cannot
                        be empty",
Toast.LENGTH_SHORT).show();
                    return;
                }

                mAuth.signInWithEmailAndPassword(email,
pass).addOnCompleteListener(new OnCompleteListener<AuthResult>() {
                    @Override
                    public void onComplete(@NonNull
                        Task<AuthResult> task) { if
                            (task.isSuccessful()) {
                                Toast.makeText(MainActivity.this, "Correct",
                                    Toast.LENGTH_SHORT).show(); Intent intent = new
                                    Intent(MainActivity.this, Personal_Details.class);
                                    startActivity(intent);
                            } else {
                                Toast.makeText(MainActivity.this, "Incorrect",
                                    Toast.LENGTH_SHORT).show();
                            }
                        }
                    }
                });
            }
        }
    );
}

```

activity_personal_details.xml;

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Personal_Details"

```

```
android:orientation="vertical"  
android:layout_margin="100px">
```

```

<TextView
    android:layout_width="match
    _parent"
    android:layout_height="wrap
    _content"
    android:text="Name:
    Navya"/>
<TextView
    android:layout_width="match
    _parent"
    android:layout_height="wrap
    _content"
    android:text="Age: 21"/>
<TextView
    android:layout_width="match
    _parent"
    android:layout_height="wrap
    _content"
    android:text="(you can add more personal details if required)"/>
<TextView
    android:id="@+id/
    btnback"
    android:layout_width="match
    _parent"
    android:layout_height="wrap
    _content" android:text="Go
    Back"
    android:layout_marginTop="1
    00px"
    android:textColor="#FF0000"
    />
</LinearLayout>

```

Personal_Details.java

```

package com.example.prg15;

import
android.content.Intent;
import
android.os.Bundle;
import
android.view.View;
import
android.widget.TextView;
import
androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class Personal_Details extends
    AppCompatActivity { @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_personal_details);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main),
            (v, insets) -> { Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
            systemBars.bottom); return insets;
        });

        TextView btnback = findViewById(R.id.btnback);
        btnback.setOnClickListener(new
            View.OnClickListener() {

```

```
@Override
public void onClick(View view) {
    Intent intent2 = new Intent(Personal_Details.this,
    MainActivity.class); startActivity(intent2);
    finish(); // Prevents multiple screen stacking instances
```

```

}
});
}
}

```

build.gradle(project) :

```

buildscript
{
    repositories {
        google()
        mavenCentral()
    }
    dependencies {
        classpath
        'com.android.tools.build:gradle:
        7.2.2' classpath
        'com.google.gms:google-
        services:4.3.15'
    }
}

```

```

task clean(type:
    Delete) { delete
    rootProject.buildDir
}

```

build.gradle (app):

```

dependencies {

    implementation
    'androidx.appcompat:appcompat:1.4.2'
    implementation
    'androidx.constraintlayout:constraintlayout:2.1.4'
    ' implementation
    'androidx.activity:activity:1.8.2'

    implementation
    platform('com.google.firebase:firebase-bom:33.1.0')
    implementation 'com.google.firebase:firebase-auth'

    testImplementation 'junit:junit:4.13.2'
    androidTestImplementation
    'com.android.support.test:runner:1.1.3'
    androidTestImplementation
    'com.android.support.test.espresso:espresso-core:3.4.0'
}

apply plugin:'com.google.gms.google-services'

```

gradle.properties File:

```

android.useAndroidX=true
android.enableJetifier=true

```


